In [1]:
```
#Data Analytics with Python Course End Project
#Insurance Data Analysis
```

In [3]:
```
#1.Import libraries such as Pandas, matplotlib, NumPy, and seaborn and load the
```

In [43]:
```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import seaborn as sns
data = pd.read_csv("insurance.csv")
```

In [7]:
```
#2.Check the shape of the data along with the data types of the column
```

In [9]:
```
data.shape
```

Out[9]:
```
(1338, 7)
```

In [13]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1338 entries, 0 to 1337
Data columns (total 7 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   age       1338 non-null   int64
 1   sex       1338 non-null   object
 2   bmi       1338 non-null   float64
 3   children  1338 non-null   int64
 4   smoker    1338 non-null   object
 5   region    1338 non-null   object
 6   charges   1338 non-null   float64
dtypes: float64(2), int64(2), object(3)
memory usage: 73.3+ KB
```

In [15]:
```
#3.Check missing values in the dataset and find the appropriate measures to fill
```

In [17]:
```
data.isna().sum()
```

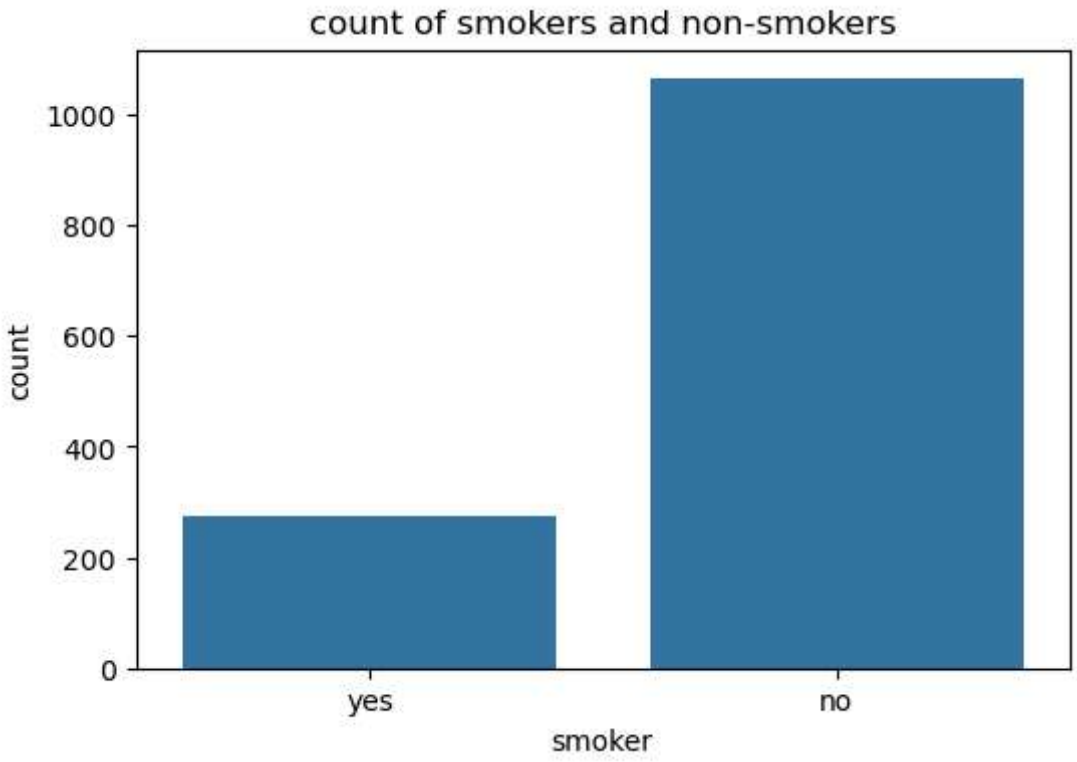
Out[17]:
```
age         0
sex         0
bmi         0
children    0
smoker      0
region      0
charges     0
dtype: int64
```

In [19]:
```
#4.Explore the relationship between the feature and target
#column using a count plot of categorical columns and a scatter plot of numerica
```

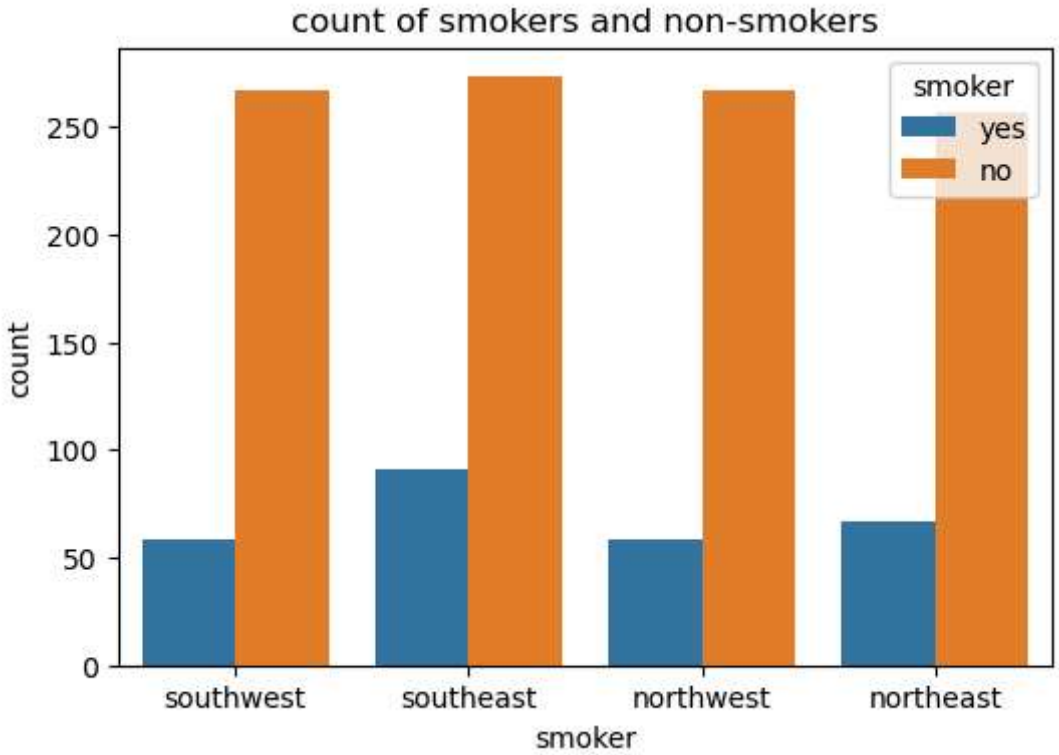
In [21]:
```
#count plot for categorical columns ; smoker,sex---charges
```

In [23]:
```
plt.figure(figsize=(6,4))
sns.countplot(x="smoker",data=data)
plt.title("count of smokers and non-smokers")
plt.xlabel("smoker")
```

```
plt.ylabel("count")
plt.show()
```

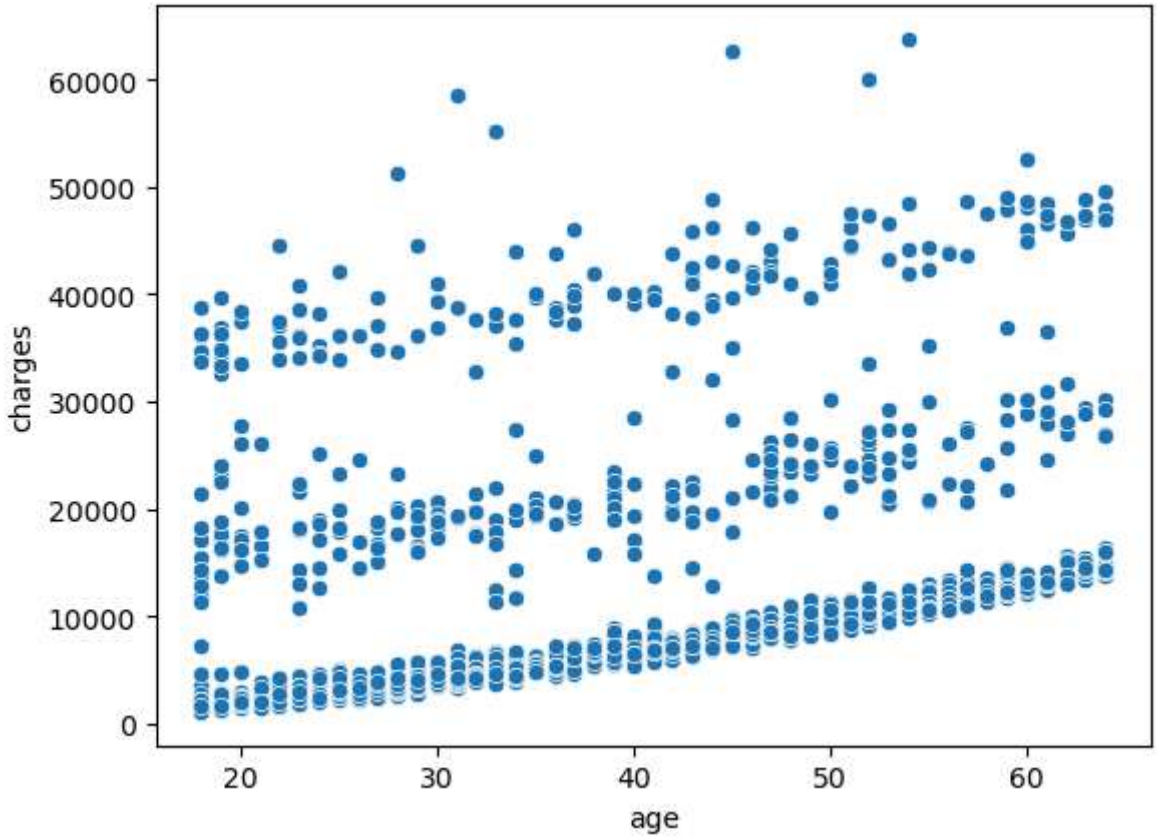


In [26]:
```
plt.figure(figsize=(6,4))
sns.countplot(x="region",data=data,hue="smoker")
plt.title("count of smokers and non-smokers")
plt.xlabel("smoker")
plt.ylabel("count")
plt.show()
```



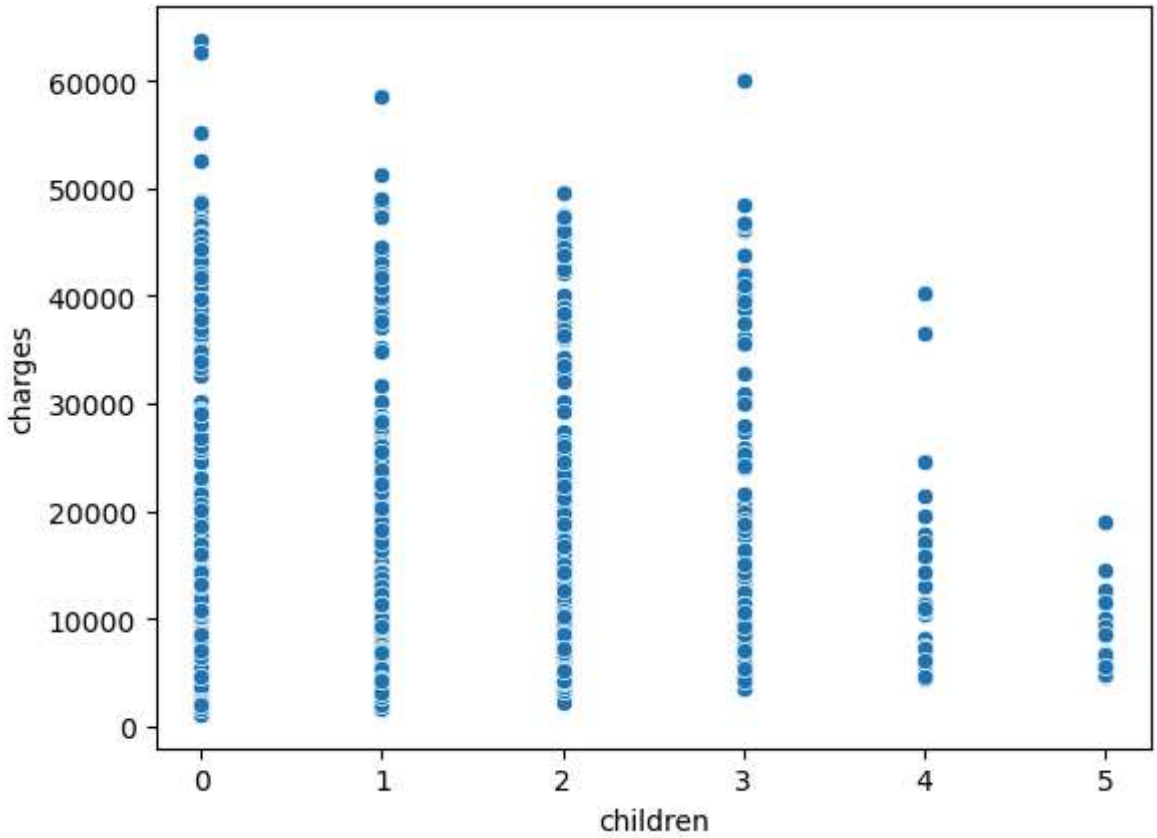
In [28]:
```
sns.scatterplot(x="age", y="charges",data=data)
```

Out[28]: <Axes: xlabel='age', ylabel='charges'>



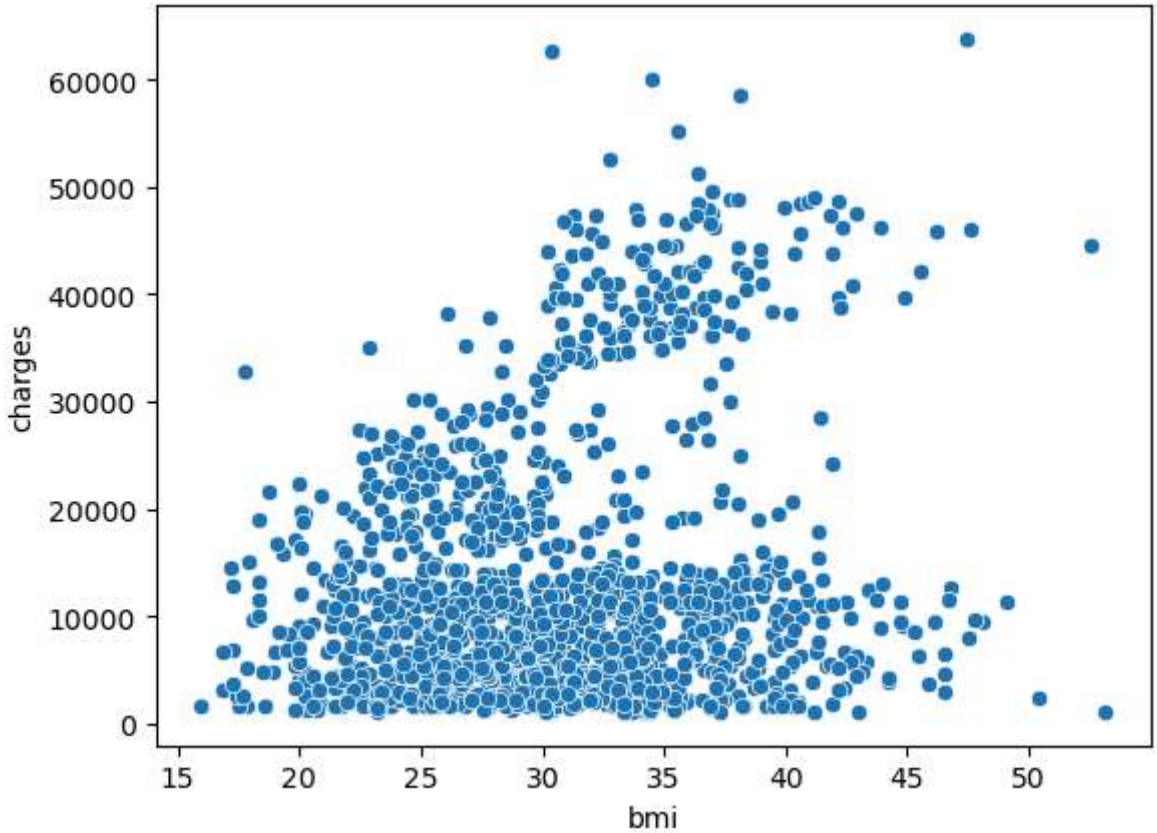
In [30]:
```
sns.scatterplot(x="children", y="charges",data=data)
```

Out[30]: <Axes: xlabel='children', ylabel='charges'>



In [37]:
```
sns.scatterplot(x="bmi", y="charges",data=data)
```

Out[37]: <Axes: xlabel='bmi', ylabel='charges'>



Age: There's a noticeable upward trend-charges tend to increase with age, especially for older individuals.

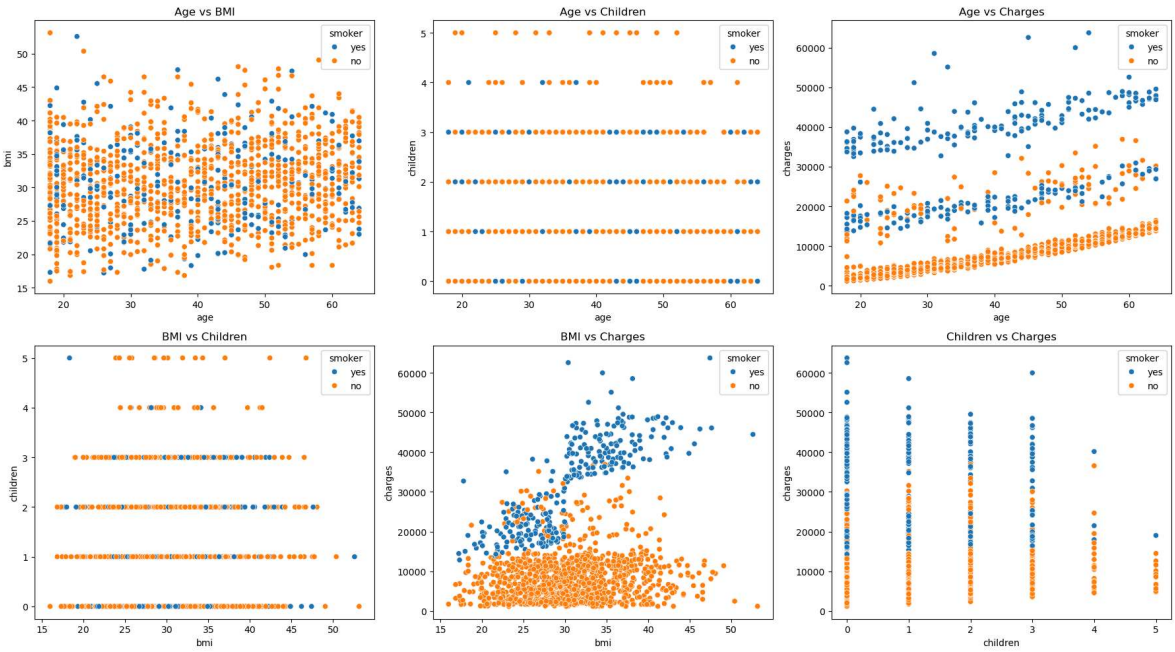
Children: No clear correlation between number of children and charges.

BMI: A slight upward trend can be seen, but it's not very strong. Some high charges appear at higher BMI levels.

In [39]:
```
#5.Perform data visualization using plots of feature vs feature
```

In [41]:
```
fig, axs = plt.subplots(2, 3, figsize=(18, 10))

sns.scatterplot(x="age", y="bmi", hue="smoker", data=data, ax=axs[0, 0])
axs[0, 0].set_title("Age vs BMI")

sns.scatterplot(x="age", y="children", hue="smoker", data=data, ax=axs[0, 1])
axs[0, 1].set_title("Age vs Children")

sns.scatterplot(x="age", y="charges", hue="smoker", data=data, ax=axs[0, 2])
axs[0, 2].set_title("Age vs Charges")

sns.scatterplot(x="bmi", y="children", hue="smoker", data=data, ax=axs[1, 0])
axs[1, 0].set_title("BMI vs Children")

sns.scatterplot(x="bmi", y="charges", hue="smoker", data=data, ax=axs[1, 1])
axs[1, 1].set_title("BMI vs Charges")

sns.scatterplot(x="children", y="charges", hue="smoker", data=data, ax=axs[1, 2]
axs[1, 2].set_title("Children vs Charges")
```

```
plt.tight_layout()
```



Age vs Charges: Charges increase with age, especially for smokers. A clear upward trend is visible for smokers, indicating a strong age- related risk factor.

BMI vs Charges: Some smokers with high BMI are charged significantly more, but the correlation is weaker than with age.

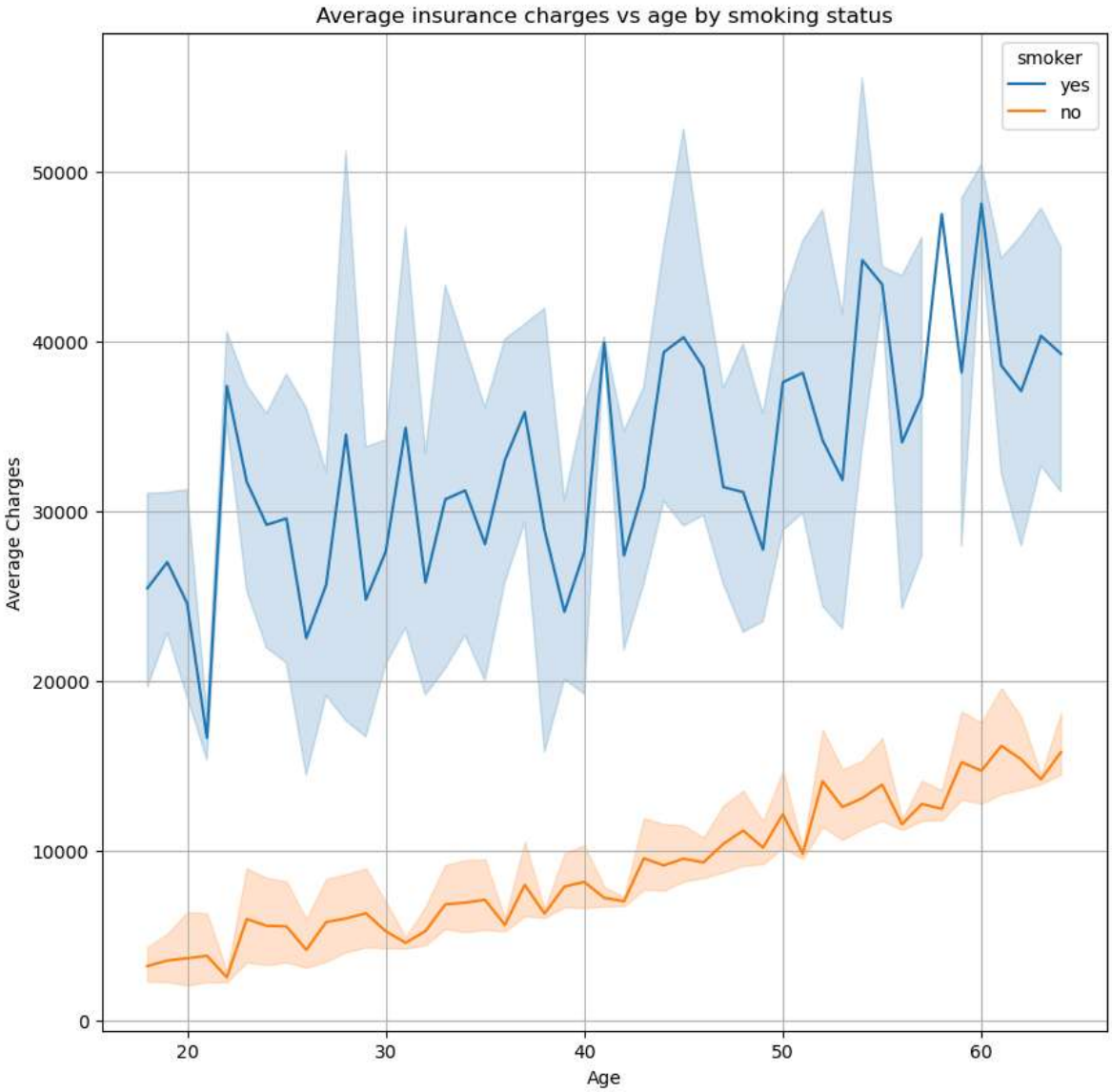
Children vs Charges: No clear pattern; the number of children has little visible effect on charges.

Age vs BMI / Children: No strong relationship-BMI and number of children appear fairly independent of age.

BMI vs Children: Again, little to no trend here.

In [46]:
```
#6.Check if the number of premium charges for smokers or non-smokers is increasi
```

In [48]:
```
plt.figure(figsize=(10,10))
sns.lineplot(data=data,x="age", y ="charges" , hue="smoker",estimator="mean")
plt.title("Average insurance charges vs age by smoking status")
plt.xlabel("Age")
plt.ylabel("Average Charges")
plt.grid(True)
plt.show()
```



Observation: Smokers: Charges rise steeply with age, especially after age 30. This suggests an accelerating cost with age for smokers.

Non-Smokers: Charges also increase with age, but at a much slower rate and more gradually.

This confirms a strong age-related rise in premium costs, particularly pronounced for smokers.